Do Now · retrieval

Synoptic retrieval: data structures

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A three-element array is accessed at index 3. What happens?

Q2

What is the worst-case Big O of a sequential search?

Q3

Push A, B, C. Which pops first?

Q4

In five buckets, h(k)=k MOD 5. Where does 12 start?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Retrieve operations — A stack pops its newest item; a queue dequeues its oldest item.

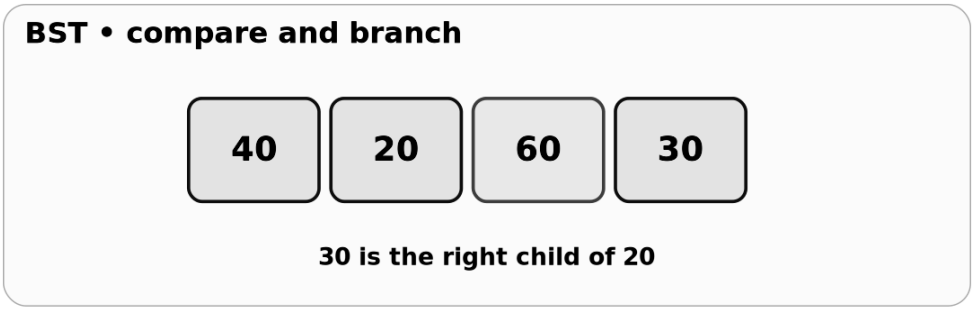
Core idea 2

Trace changing state — After Push 3, Push 6, Pop, Push 9, the stack is [3,9]. After Enqueue A,B then Dequeue, the queue is [B].

Core idea 3

Apply the BST rule — Insert 40 first, 20 left, 60 right, then 30 as the right child of 20.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LIFO — Last In, First Out.

FIFO — First In, First Out.

head — Reference to the first node in a linked list.

BST — Binary search tree ordered by key comparisons.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

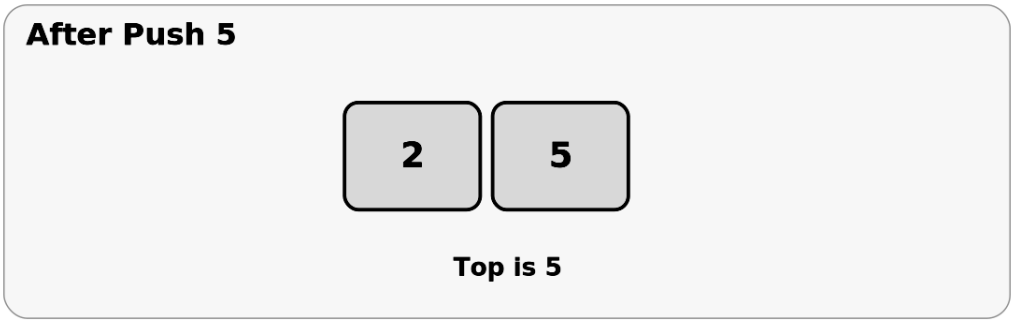
Problem

Trace stack operations Push(2), Push(7), Pop(), Push(5).

Follow the worked method

1. Predict first — Start with [2,7]. The top of the stack is 7.
2. Check the method — Pop removes 7, leaving [2].
3. Check the method — Push 5 gives [2,5]; the top is now 5.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
After the two enqueues, the queue is [X,Y].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A stack contains [4,9] bottom-to-top. Push 2, Pop, Pop. State both removed values and remaining stack.

Question 2

A queue receives 5,8,3. It dequeues once then enqueues 2. Give the queue and explain why its next dequeue removes 8.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Insert 40,20,60,10,30,25 into a BST. Give the parent of 25, state whether it is a left/right child, and explain what happens to worst-case height when inputs arrive sorted.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A queue normally removes the most recently enqueued item.

Use key vocabulary from memory

1. Explain LIFO in your own words.

2. Explain FIFO in your own words.

3. Explain head in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.